



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	FIN6131
Course title (English)	Portfolio and Asset Management
Course title (Chinese)	投资组合和资产管理
Units	3
Language of Instruction	English
Description (English)	This course offers a comprehensive overview of modern portfolio management, integrating both classical financial theories and contemporary quantitative techniques. Students will learn essential concepts such as portfolio theory, risk and return trade-offs, asset pricing models, factor investing, and optimal portfolio construction. The curriculum further explores advanced topics including covariance matrix estimation and Bayesian shrinkage, Black-Litterman model, volatility modeling, bootstrap methods, behavioral finance, and the application of machine learning in asset allocation. Through practical projects, students will develop the ability to construct, analyze, and evaluate investment portfolios using data-driven approaches and modern analytical tools.
Description (Chinese)	本课程通过结合经典金融理论与当代量化技术，全面介绍现代投资组合管理。学生将学习投资组合理论、风险与收益权衡、资产定价模型、因子投资以及最优投资组合构建等核心概念。课程还将深入探讨协方差矩阵估计与贝叶斯收缩、Black-Litterman模型、波动率建模、自助法、行为金融以及机器学习在资产配置中的应用。通过实践项目，学生将掌握利用数据驱动方法和现代分析工具构建、分析及评估投资组合的能力。

2. Prerequisites / Co-requisites

A. Prerequisites

Undergraduate level knowledge of probability, statistics, linear algebra, and econometrics.

Familiarity with Python (assignments will require coding with Python).

Basic knowledge of finance and investment is beneficial.



B. Co-requisites

None

3. Learning Outcomes

Knowledge Outcomes:

- Understand the foundational theories of modern portfolio management, including risk-return trade-offs, asset pricing models, and factor investing.
- Master techniques for asset allocation, optimal portfolio construction, and evaluation of investment performance.
- Grasp advanced topics such as covariance matrix estimation, denoising methods (e.g., random matrix theory), Bayesian shrinkage, and the Black-Litterman model.
- Comprehend the impact of behavioral finance and on portfolio decisions.
- Acquire practical knowledge in applying quantitative and statistical methods to real-world portfolio management problems.

Skills Outcomes:

- Apply Python and quantitative tools to implement portfolio optimization, risk analysis, and performance evaluation.
- Analyze and interpret financial data using econometric and statistical methods.
- Construct, backtest, and refine investment portfolios based on data-driven approaches.
- Communicate portfolio analysis results and investment strategies effectively to both technical and non-technical audiences.

Values Outcomes:

- Develop a rigorous, analytical, and data-driven mindset for portfolio management and investment decision-making.
- Appreciate the integration of financial theory, quantitative analysis, and practical implementation in asset management.
- Cultivate critical awareness of model limitations, estimation errors, and the assumptions underlying portfolio models.
- Maintain objectivity and integrity in portfolio research, being mindful of issues such as overfitting, data snooping, and behavioral biases.



4. Course syllabus

Please refer to Indicative teaching plan.

5. Assessment Scheme

Component/ method	% weight
Assignments	20
Mid-term Exam	40
Final Project	40

6. Grade descriptor

General

Grade	Description
A	Outstanding understanding of the theory and methodology of modern portfolio management and asset allocation, with exceptional application of quantitative and statistical techniques to real-world financial data using Python. Demonstrates advanced ability to construct, analyze, and optimize investment portfolios, accurately interpret risk and return dynamics, and critically evaluate empirical results. Strongly exceeds expectations for high-quality research in empirical asset management and for professional roles in the finance industry.
A-	Good understanding of the theory and methodology of modern portfolio management and asset allocation, with exceptional application of quantitative and statistical techniques to real-world financial data using Python. Demonstrates a high level of ability to construct, analyze, and optimize investment portfolios, accurately interpret risk and return dynamics, and critically evaluate empirical results, exceeding expectations for high-quality research in empirical asset management and for professional roles in the finance industry.
B+, B, B-	Satisfactory understanding of the theory and methodology of modern portfolio management and asset allocation, with exceptional application of quantitative and statistical techniques to real-world financial data using Python. Demonstrates adequate ability to construct, analyze, and optimize investment portfolios, accurately interpret risk and return dynamics, and critically evaluate empirical results, meeting or slightly exceeding expectations for high-quality research in empirical asset management and for professional roles in the finance industry.
C+, C, C-	Basic understanding of the theory and methodology of modern portfolio management and asset allocation, with exceptional application of quantitative and statistical techniques to real-world financial data using Python. Demonstrates basic



	ability to construct, analyze, and optimize investment portfolios, accurately interpret risk and return dynamics, and critically evaluate empirical results, meeting minimum requirements for research in empirical asset management and for professional roles in the finance industry.
D+, D	Limited understanding of the theory and methodology of modern portfolio management and asset allocation, with exceptional application of quantitative and statistical techniques to real-world financial data using Python. Lacks sufficient ability to construct, analyze, and optimize investment portfolios to meet the standards for research or industry expectations.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

CTE;

Communicate with instructor and TAs in class, via email, during office hours, etc.

8. Reading

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Course Introduction & Asset Management Overview
2	Portfolio Theory & Risk-Adjusted Performance Metrics
3	Asset Pricing & Factor Models
4	Barra Factor Model
5	Covariance Estimation & Bayesian Shrinkage
6	Common Asset Allocation Methods & Kelly Formula
7	Black-Litterman Model
8	Midterm (tentative)
9	Volatility Modeling
10	Bootstrap Methods in Portfolio Analysis
11	Behavioral Finance in Portfolio Management



12	Machine Learning for Asset Allocation
13	Tail Risks and Extreme Value Modeling
14	Final Presentation

11. Any other information

12. Version date

Version number	1
As of (date)	2025/07/22